CSE400 Fundamentals of Probability in Computing

Finance Group 1 Members:

- Kush Kelaiya AU2440018
- Kesar Thakkar AU2420041
- Ishika Patel AU2440088
- Pranav Jain AU2420166
- Nissarg Danidhariya AU2440261

Project Context and Motivation

Project Title:

Algorithmic Portfolio Optimization

Application Domain:

Finance and Stock Market

Project Context:

- The project focuses on modeling uncertainty in stock market returns.
- Stock market depends on market conditions.
- Due to these reasons, stock market returns are treated as random variables.

Project Motivation:

- This document explains that how probability is used in optimization of portfolio.
- It connects stock market returns, risk and uncertainty.
- The methods used in the document align well with the probability concepts that we are studying or will study in this course.

Project Objective and Scope

Project Objective:

- To study uncertainty in stock market returns using probability.
- To model stock market returns as random variables.
- To analyze portfolio risk and expected return under uncertainty.
- To understand the effect of asset dependency on the performance of portfolio.

Project Scope:

- Using historical data of stock price from the document.
- Applying Probability Distributions to model returns of stock market.
- Evaluation of portfolio performance using risk and return.
- Using Covariance to represent asset dependency.

Project System Overview

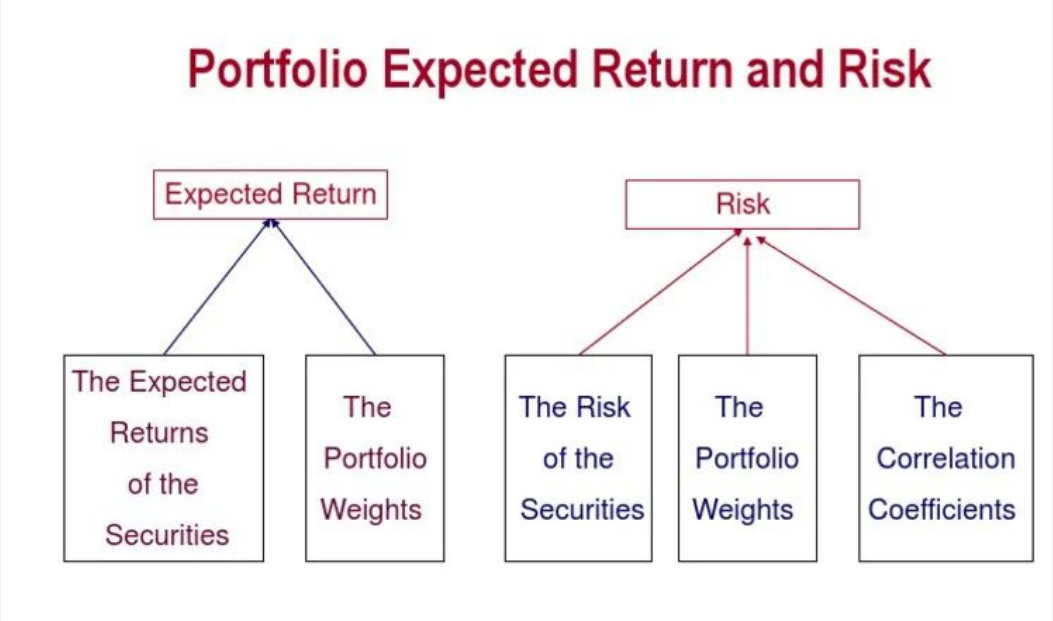
- This system studies a set of financial assets (stock).
- The data of historical stock price is collected for these financial assets.
- The stock prices are then converted into stock returns.
- These stock returns are treated as random variables.
- Probability distributions are fitted to model the uncertainty of returns.
- Dependence between stock returns is modeled using covariance matrix.
- Portfolio return is calculated using weights of assets.
- Portfolio risk is calculated using variance.
- Different portfolios are generated, and then comparison among them is done.
- This system helps in identifying portfolios which have better risk and return balance.

Sources of Uncertainty in the Project

- Stock prices fluctuate due to market behaviour.
- The economic conditions affects the stock returns.
- Market volatility causes uncertainty in daily stock returns.
- Sudden movements in prices increases the risk of return.
- Correlation between asset returns results in changes in portfolio risk.
- Return estimation depends on the historical data.
- Historical data might not be accurate in predicting the future returns.
- Returns of different stocks are correlated with each other.
- Individual stock returns varies randomly over time.

Key Random Variables and Their Roles

- Individual stock returns are treated as random variables.
- Portfolio return is a random variable which is formed from stock returns.
- Stock returns represents uncertainty in the performance of assets.
- Portfolio risk is measured using variance.
- Portfolio return shows the overall profit or loss.
- The covariance matrix models dependence between the stock returns.
- The dependency on assets affects the portfolio risk.



Probabilistic Models and Assumptions

- The returns on stock are modeled by using probability distributions.
- Historical data of stock returns is used to estimate these distributions.
- Portfolio risk is modeled by using covariance matrix.
- The covariance matrix captures dependence between the stock and returns.
- Stock returns are assumed to be independent of time.
- Dependence between the assets are assumed to remain constant.
- Historical data is used to predict future uncertainty.

Probabilistic Reasoning and Inference Logic

How uncertainty propagates through the system

- Asset returns are modeled as random variables
- Portfolio return is a weighted combination of uncertain returns
- Variance and covariance determine how risk accumulates or diversifies

How probabilistic reasoning supports inference / decision-making

- Probability distributions summarize possible outcomes
- Expected value captures average performance
- Variance and tail probabilities capture risk exposure
- Decisions are based on likelihood, not certainty

Step-by-step conceptual reasoning pathway

1. Model asset returns as random variables
2. Use historical data to estimate distributions
3. Combine assets to form portfolio outcomes
4. Infer risk–return trade-offs probabilistically

Operationalizing Probabilistic Reasoning in the Project

How Slide 7 reasoning is realized

- Simulate multiple portfolios using random weight assignments
- Compute portfolio return and risk from sampled outcomes
- Observe empirical distributions from simulations

Evaluation metrics & probabilistic meaning

- Expected return → mean of portfolio distribution
- Volatility → standard deviation (risk measure)
- Sharpe ratio → risk-adjusted expected performance

Simplifying assumptions (for feasibility)

- Asset returns are stationary over time
- Risk-free rate assumed constant
- No transaction costs or market impact

Current Limitations and Conceptual Gaps

Known weaknesses

- Historical data may not reflect future behavior
- Extreme market events are underrepresented

Unverified / questionable assumptions

- Returns follow stable distributions
- Asset correlations remain constant

Conceptual ambiguities

- How well simulations approximate real markets
- Sensitivity to distributional assumptions

Planned Refinements and Role Coordination

What evolves in next milestone

- Formal mathematical modeling of returns and risk
- Clear definition of random variables and distributions

Upcoming conceptual / modeling tasks

- Define joint distributions for assets
- Analyze impact of covariance on portfolio risk
- Introduce optimization criteria

Role of the Reviewer / GitHub Manager

- Maintain a centralized and version-controlled project repository
- Ensure consistency across code, reports, and documentation
- Review contributions for conceptual alignment with project goals

Summary of Group-Level Understanding

- A unified view of how uncertainty, randomness, and inference shape the problem
- Shared understanding of probabilistic modeling as the foundation for risk analysis
- Clear alignment on how probability supports decision-making under uncertainty